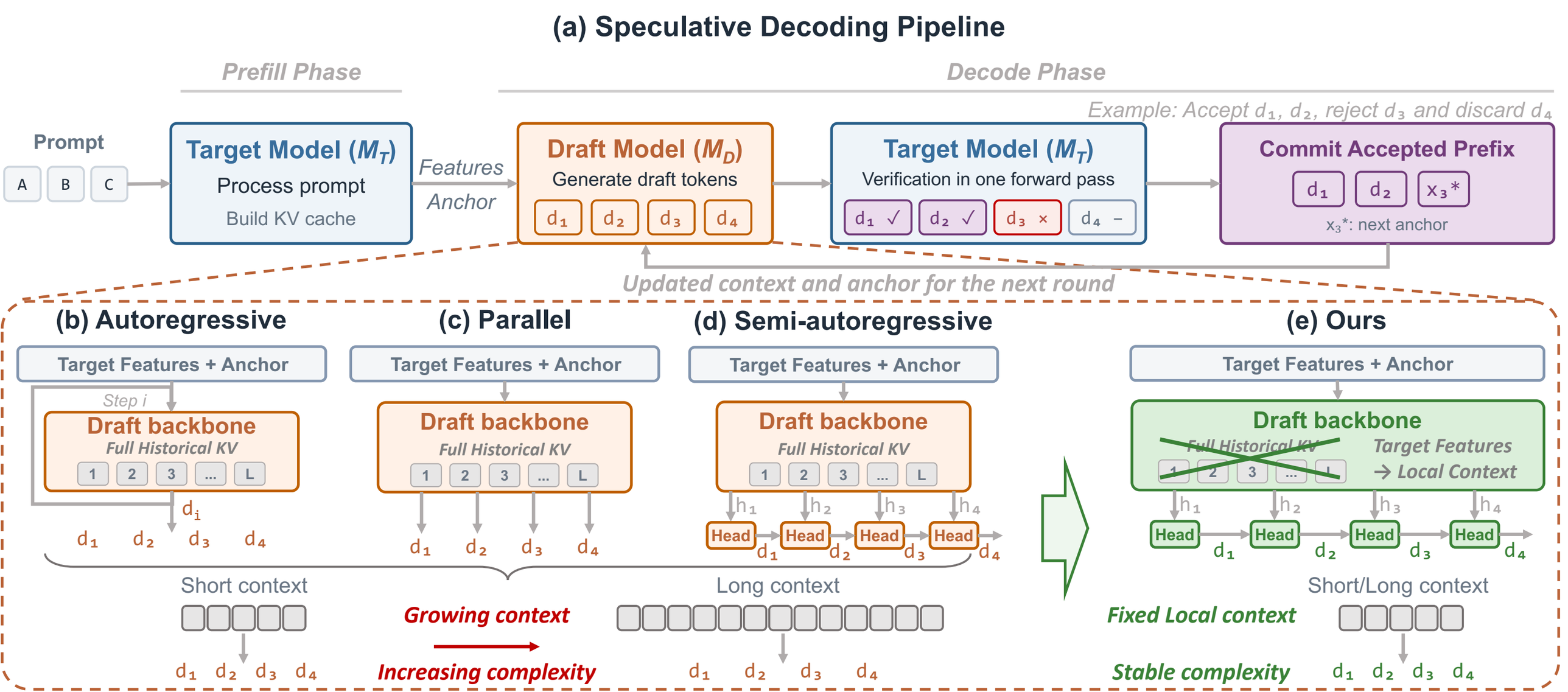

(a) Speculative Decoding Pipeline
Prefill Phase
Decode Phase
Example: Accept d_1, d_2, reject d_3 and discard d_4
Prompt
A
B
C
Target Model (M_T)
Process prompt
Build KV cache
Features
Anchor
Draft Model (M_D)
Generate draft tokens
d_1 d_2 d_3 d_4
Target Model (M_T)
Verification in one forward pass
d_1 ✓ d_2 ✓ d_3 × d_4 –
Commit Accepted Prefix
d_1 d_2 x_3*
x_3*: next anchor
Updated context and anchor for the next round
(b) Autoregressive
Target Features + Anchor
Step i
Draft backbone
Full Historical KV
1 2 3 ... L
d_i
d_1 d_2 d_3 d_4
Short context
d_1 d_2 d_3 d_4
(c) Parallel
Target Features + Anchor
Draft backbone
Full Historical KV
1 2 3 ... L
d_1 d_2 d_3 d_4
Long context
d_1 d_2 d_3 d_4
Growing context
Increasing complexity
(d) Semi-autoregressive
Target Features + Anchor
Draft backbone
Full Historical KV
1 2 3 ... L
h_1 h_2 h_3 h_4
Head Head Head Head
d_1 d_2 d_3 d_4
(e) Ours
Target Features + Anchor
Draft backbone
Full Historical KV
1 2 3 ... L
Target Features
→ Local Context
h_1 h_2 h_3 h_4
Head Head Head Head
d_1 d_2 d_3 d_4
Short/Long context
d_1 d_2 d_3 d_4
Fixed Local context
Stable complexity